

Question 1

$$a) \quad \mu = \frac{345 + 420 + 390 + 410 + 380 + 400 + 430 + 360 + 375 + 390 + 405 + 415 + 400 + 385 + 395}{15}$$

$$= \frac{5900}{15}$$

$$= 393.33$$

The point estimate of $\mu = \text{RM } 393.33$

$$b) \quad s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}}$$

$$= \sqrt{\frac{(345-393.33)^2 + (420-393.33)^2 + \dots + (395-393.33)^2}{15-1}}$$

$$= 22.49$$

Degree of freedom = $15-1$

$$= 14$$

$$t_{0.95, 14} = 2.145$$

$$95\% \text{ confidence level} = 393.33 \pm 2.145 \left(\frac{22.49}{\sqrt{15}} \right)$$

$$= (380.87, 405.79)$$

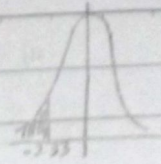
We are 95% confident that the average monthly grocery expenditures is between 380.87 and 405.79.

No.:

Date:

Question 2

a. $H_0: \mu = 5$
 $H_1: \mu < 5$

d. Reject H_0 if $Z_{test} < -2.33$.Since $Z_{test} = -7.45 (< -2.33)$, H_0 is rejected

b. $Z_{test} = \frac{4.6 - 5}{\frac{1.2}{\sqrt{500}}}$
 $= -7.45$

e. There is sufficient evidence to conclude that at $\alpha = 0.01$, the average waiting time for customers in the bank's queue is less than 5 minutes.

c. $\alpha = 0.01$

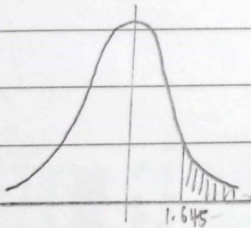
$Z_{0.01} = 2.33$

Question 3

$H_0: \mu = 160$

$H_1: \mu > 160$

$Z_{test} = \frac{183 - 160}{\frac{12}{\sqrt{25}}}$
 $= 9.58$



$\alpha = 0.05$

$Z_{0.05} = 1.645$

Reject H_0 if $Z_{test} > 1.645$.

Since $Z_{test} = 9.58 (> 1.645)$, H_0 is rejected.

Thus, at $\alpha = 0.05$, there is sufficient evidence to conclude that the mean score for the student from this university is greater than 160.

Question 4

$H_0: \mu = 50$

$H_1: \mu > 50$

$Z_{test} = \frac{51 - 50}{\frac{3.6}{\sqrt{38}}}$
 $= 1.71$

$\alpha = 0.01$

$Z_{0.01} = 2.33$

Reject H_0 if $Z_{test} > 2.33$.

Since $Z_{test} = 1.71 (< 2.33)$, H_0 is not rejected.

Thus, at $\alpha = 0.01$, there is insufficient evidence to conclude that the population mean is greater than 50cm.

Question 5

$$H_0: \sigma = 7460$$

$$H_1: \sigma > 7460$$

$$\chi^2 = \frac{(15-1)(7871)^2}{7460^2}$$

$$= 55.661$$

$$\alpha = 0.01$$

$$\chi^2_{0.01, 50} = 76.154$$

Reject H_0 if $\chi^2 > 76.154$.

Since $\chi^2 = 55.661 (< 76.154)$, H_0 is not rejected.

Thus, at $\alpha = 0.01$, there is sufficient evidence to conclude that the males have a standard deviation that is greater than the females.

Question 6

a. Female:

$$\bar{x}_1 = \frac{46.4}{12}$$

$$s_1 = 0.563$$

$$= 3.867$$

Male:

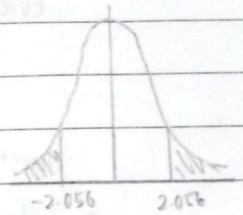
$$\bar{x}_2 = \frac{64.3}{16}$$

$$s_2 = 0.395$$

$$= 4.019$$

b. $H_0: \mu_1 - \mu_2 = 0$

$$H_1: \mu_1 \neq \mu_2$$



$$\alpha = 0.05 \quad df = 12 + 16 - 2 = 26$$

$$t_{0.025, 26} = 2.056$$

Reject H_0 if $t_0 < -2.056$ or $t_0 > 2.056$.

$$S_p = \sqrt{\frac{11(0.563)^2 + 15(0.395)^2}{12 + 16 - 2}}$$

$$= 0.47$$

$$t_0 = \frac{3.867 - 4.019 - 0}{0.47 \sqrt{\frac{1}{12} + \frac{1}{16}}}$$

$$= -0.847$$

Since $-2.056 < t_0 < 2.056$, H_0 is not rejected.

c. Thus, at $\alpha = 0.05$, there is insufficient evidence to conclude that the two samples have different mean.

$$d. H_0: \sigma_1^2 = \sigma_2^2$$

$$H_1: \sigma_1^2 > \sigma_2^2$$

$$F = \frac{0.563^2}{0.395^2}$$

$$= 2.03$$

No.: _____

Degree of freedom

$$\text{Numerator} = 12 - 1 = 11$$

$$\text{Denominator} = 16 - 1 = 15$$

$$F_{0.01, 11, 15} = 3.7$$

Reject H_0 if $F > 3.7$ Since $F = 2.03 (< 3.7)$, H_0 is not rejected.Thus, at $\alpha = 0.01$, there is insufficient evidence to conclude that $\sigma_1^2 > \sigma_2^2$.Question 7

a. Difference, D	-0.5	0	-2.5	-4.0	1.0	-1.5	0	-4.0	-1.0	3.0	$\sum D = -9.5$
$D^2 = (x_1 - x_2)^2$	0.25	0	6.25	16.0	1.0	2.25	0	16.0	1.0	9.0	$\sum D^2 = 51.75$

$$\begin{aligned}
 \text{b. } \bar{D} &= \frac{\sum D}{n} \\
 &= \frac{-9.5}{10} \\
 &= -0.95 \\
 S_D &= \sqrt{\frac{\sum D^2 - \frac{(\sum D)^2}{n}}{n-1}} \\
 &= \sqrt{\frac{51.75 - \frac{(-9.5)^2}{10}}{10-1}} \\
 &= 2.4
 \end{aligned}$$

$$\text{c. } H_0: \mu_D = 0$$

$$H_1: \mu_D \neq 0$$

$$\begin{aligned}
 t &= \frac{-0.95 - 0}{\frac{2.4}{\sqrt{10}}} \\
 &= -1.25
 \end{aligned}$$

$$\alpha = 0.05$$

$$\frac{\alpha}{2} = 0.025 \quad df = 10 - 1 = 9$$

$$T\text{-value} = 2.262$$

Reject H_0 if $t < -2.262$ or $t > 2.262$.

Since $t = -1.25 (-2.262 < t < 2.262)$, H_0 is not rejected. Thus, at $\alpha = 0.05$, there is sufficient evidence to conclude that there is no difference in heights between mothers and their first daughters.

Question 8

a.

Cell	Observed count, o_i	Expected count, e_i	$[o_i - e_i]^2 / e_i$
1, 1	90	93.02	0.098
1, 2	80	79.07	0.011
1, 3	30	27.91	0.157
2, 1	70	69.77	0.008
2, 2	60	59.30	0.008
2, 3	20	20.93	0.041
3, 1	40	37.21	0.209
3, 2	30	31.63	0.084
3, 3	10	11.16	0.121
χ^2			0.737

b. H_0 : There is no association between customer from the bank H_1 : There is association between customer... from the bank

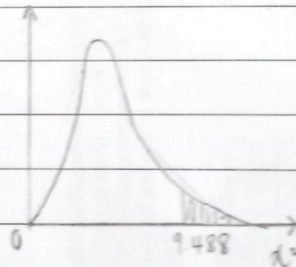
$$\alpha = 0.05$$

$$df = (3-1)(3-1)$$

$$= 4$$

$$\text{Test Statistic: } \chi^2 = 0.737$$

$$\text{Critical value: } \chi^2_{k=4, \alpha=0.05} = 9.488$$



Since $\chi^2 = 0.737 (< 9.488)$, H_0 is not rejected.

Thus, at $\alpha = 0.05$, there is insufficient evidence to conclude that there is association between customer from the bank.